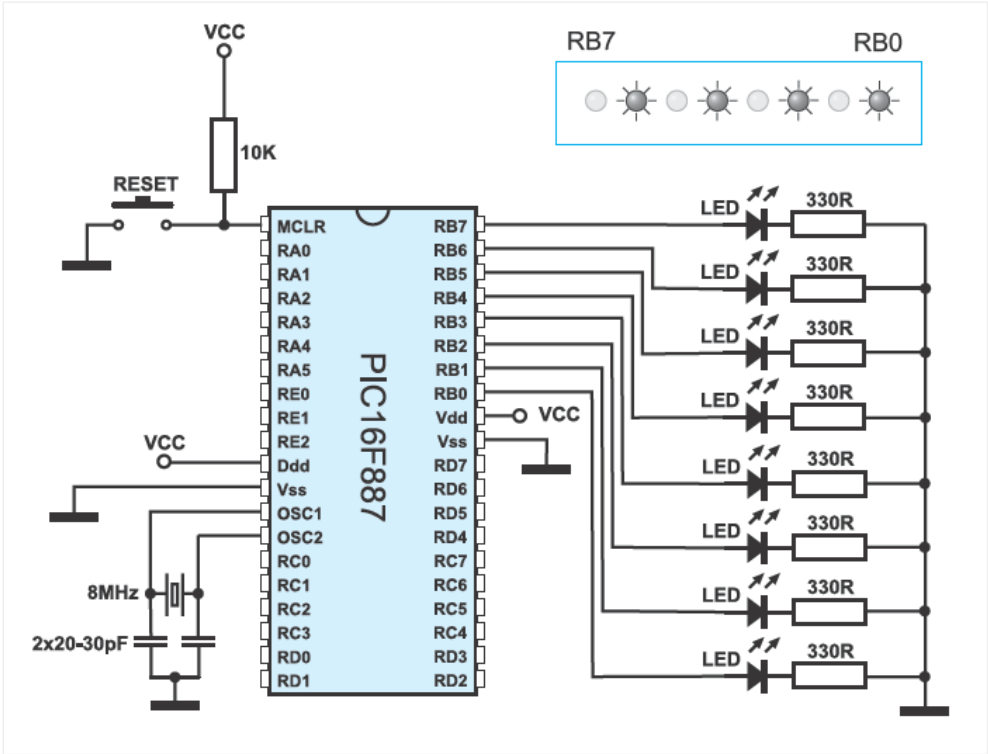


4.3 EXAMPLE 1

MIKROELEKTRONIKA

Writing header, configuring I/O pins, using delay function and switch operator

The only purpose of this program is to turn on a few LED diodes on port B. Anyway, use this example to study what a real program looks like. Figure below shows connection schematic, while the program is on the next page.



When switching on, every other LED diode on the port B emits light, which indicates that the microcontroller is properly connected and operates normally.

This example describes a correctly written header. It's the same for all the programs described in this book. To skip repetitiveness, it will not be written in the following examples, but is considered to be at the beginning of every program and marked as 'Header'.

Example 1

Header

```

/*
 * Program name:
 *   Example 1
 * Copyright:
 *   (c) MikroElektronika, 2005-2009
 * Description:
 *   This is a simple program used to demonstrate the operation of the micro-
 *   controller. Every second LED on port B is turned on.
 * Configuration:
 *   Microcontroller: PIC16F887
 *   Device:         EasyPIC5
 *   Oscillator:     HS, 08.0000 MHz
 *   SW:             mikroC PRO v8.0
 * Notes: -
 */

```

Header is placed at the beginning of the program and gives basic information in the form of comments (name of the program, release date etc.). Don't be deluded into thinking that after a few months you will know what that program is about and why you saved it.

Program execution

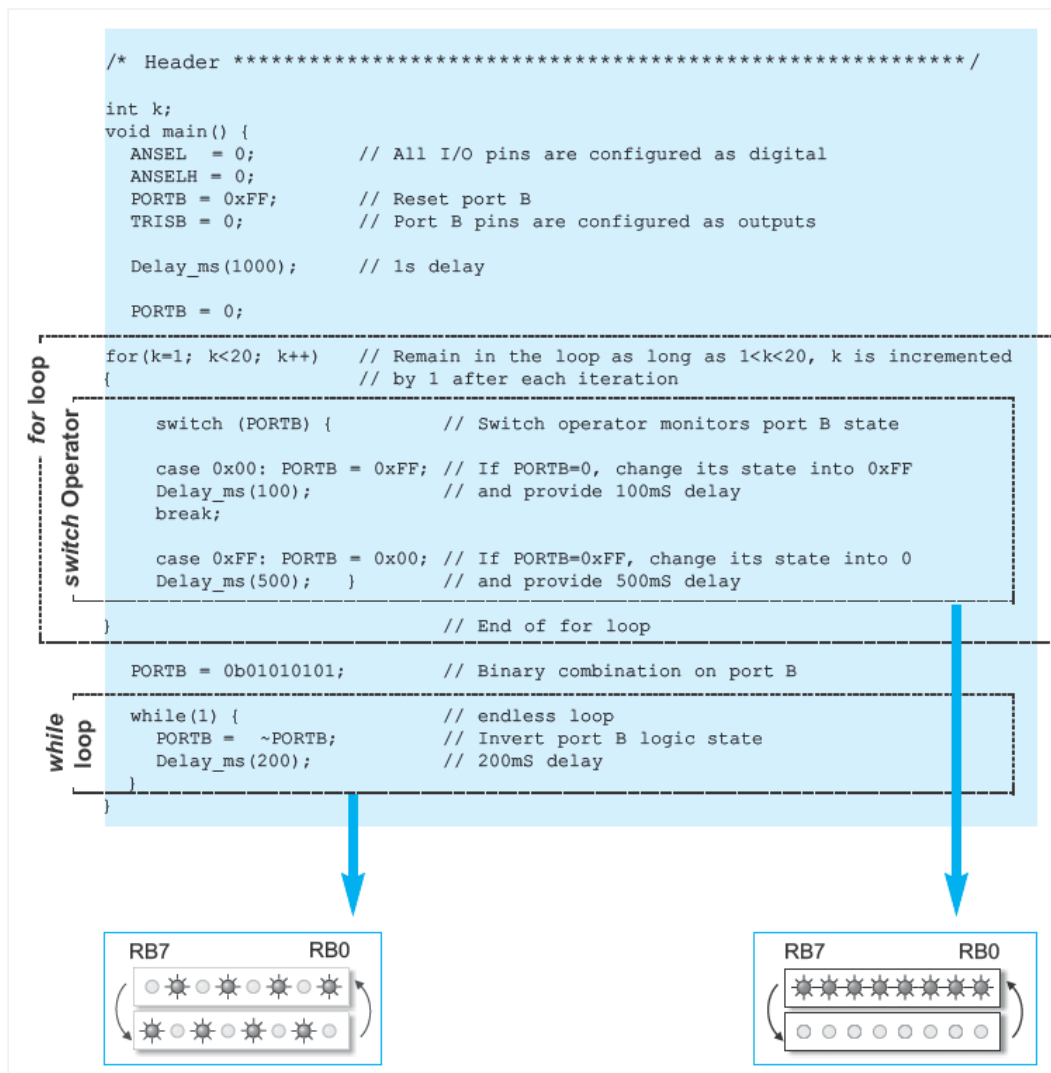
```

void main() {
    ANSEL = 0;           // All I/O are configured as digital
    ANSELH = 0;
    PORTB = 0b01010101; // Binary combination on port B
    TRISB = 0;           // Port B pins are configured as outputs
}

```

To make this example more interesting, we will enable LEDs connected to the port B to blink. There are several ways to do it:

1. As soon as the microcontroller is turned on, all LEDs will emit light for a second. The *Delay* function is in charge of it in the program. It's only needed to set delay expressed in milliseconds.
2. After one second, the program enters the *for* loop and remains there as long as the variable *k* is less than 20. The variable is incremented by 1 after each iteration. Within the *for* loop, the *switch* operation monitors port B logic state. If `PORTB=0xFF`, its state is inverted into `0x00` and vice versa. Any change of these logic states causes all LEDs to blink. Duty Cycle is 5:1 (500mS:100mS).
3. When the program exits the *for* loop, the port B logic state changes (`0xb 01010101`) and the program enters the endless *while* loop and remains there as long as `1=1`. The port B logic state is inverted each 200mS.



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